

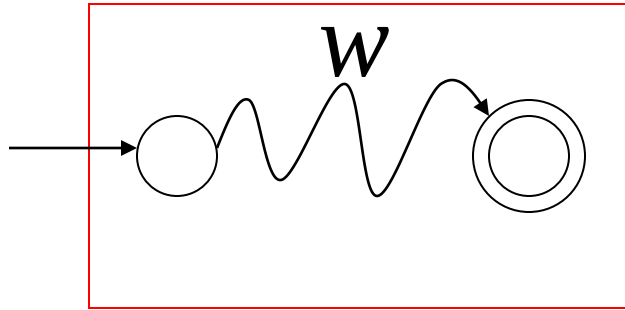
Decidable Problems of Regular Languages

Membership Question

Question: Given regular language L
and string w
how can we check if $w \in L$?

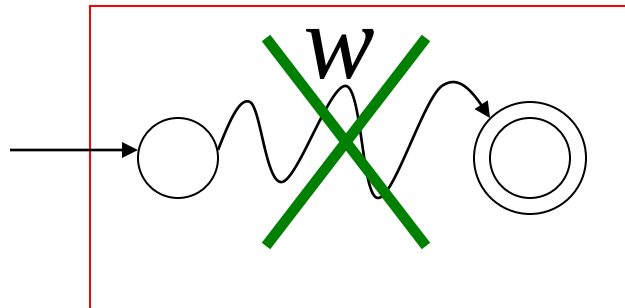
Answer: Take the DFA that accepts L
and check if w is accepted

DFA



$$w \in L$$

DFA



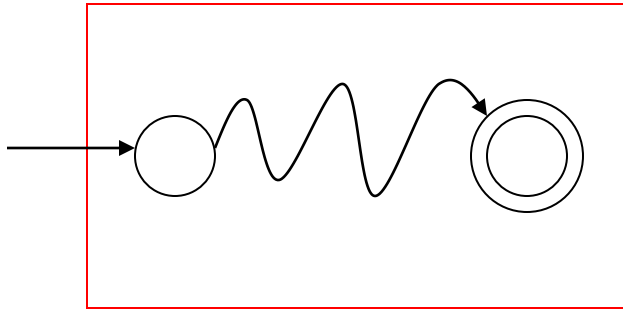
$$w \notin L$$

Question: Given regular language L
how can we check
if L is empty: $(L = \emptyset)$?

Answer: Take the DFA that accepts L

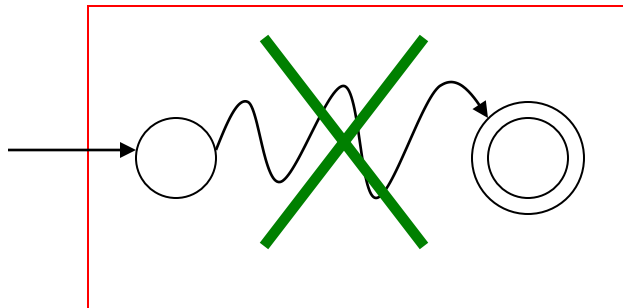
Check if there is any path from
the initial state to an accepting state

DFA



$$L \neq \emptyset$$

DFA



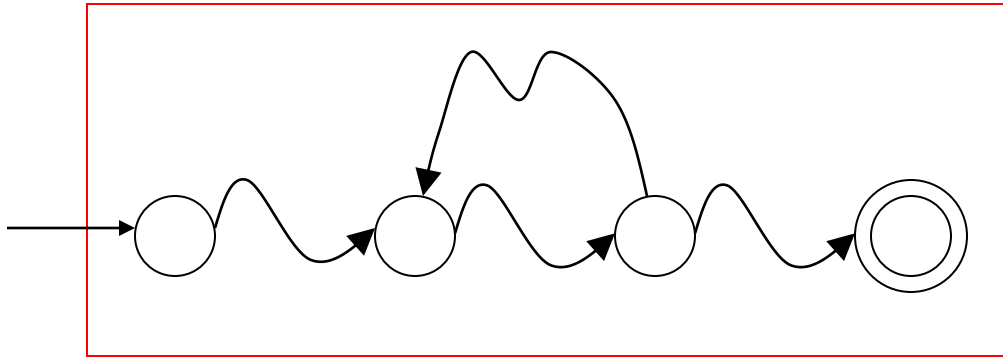
$$L = \emptyset$$

Question: Given regular language L
how can we check
if L is finite?

Answer: Take the DFA that accepts L

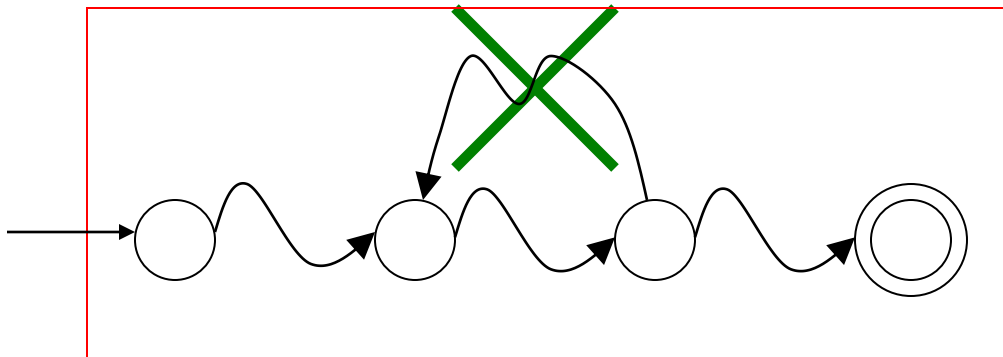
Check if there is a walk with cycle
from the initial state to a final state

DFA



L is infinite

DFA



L is finite

Question: Given regular languages L_1 and L_2
how can we check if $L_1 = L_2$?

Answer: Find if $(L_1 \cap \overline{L_2}) \cup (\overline{L_1} \cap L_2) = \emptyset$

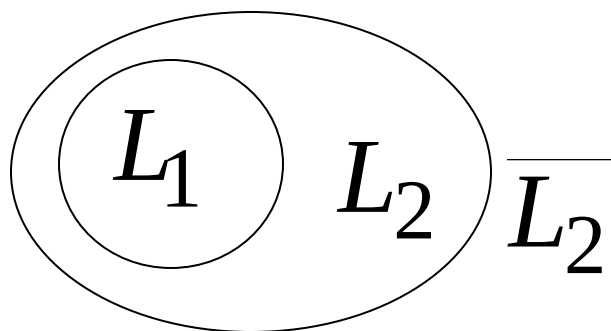
$$(L_1 \cap \overline{L_2}) \cup (\overline{L_1} \cap L_2) = \emptyset$$



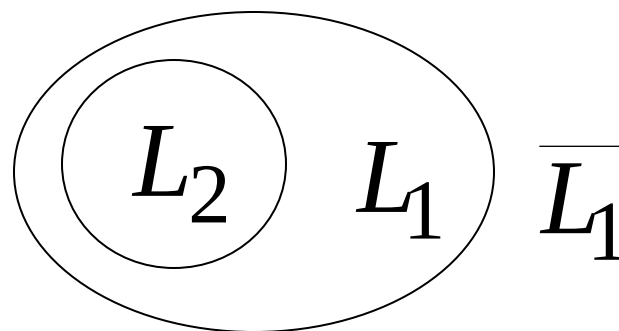
$$L_1 \cap \overline{L_2} = \emptyset$$

and

$$\overline{L_1} \cap L_2 = \emptyset$$



$$L_1 \subseteq L_2$$



$$L_2 \subseteq L_1$$



$$L_1 = L_2$$

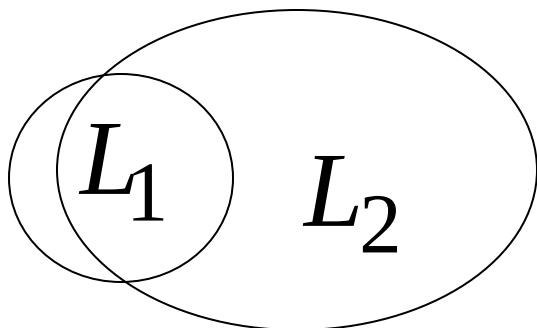
$$(L_1 \cap \overline{L_2}) \cup (\overline{L_1} \cap L_2) \neq \emptyset$$



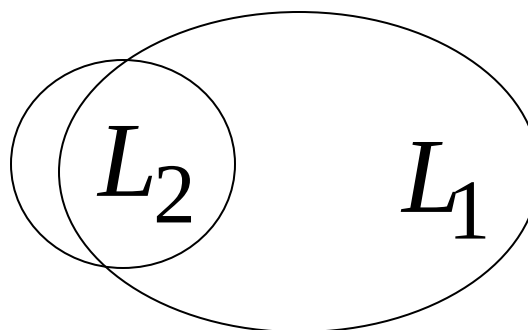
$$L_1 \cap \overline{L_2} \neq \emptyset$$

or

$$\overline{L_1} \cap L_2 \neq \emptyset$$



$$L_1 \not\subset L_2$$



$$L_2 \not\subset L_1$$



$$L_1 \neq L_2$$

Decidable Problems of Context-Free Languages

Membership Question:

for context-free grammar G
find if string $w \in L(G)$

Membership Algorithms: Parsers

- Exhaustive search parser
- **CYK** parsing algorithm

Empty Language Question:

for context-free grammar G

find if $L(G) = \emptyset$

Algorithm:

1. Remove useless variables
2. Check if start variable S is useless

Infinite Language Question:

for context-free grammar G

find if $L(G)$ is infinite

Algorithm:

1. Remove useless variables
2. Remove unit and λ productions
3. Create dependency graph for variables
4. If there is a loop in the dependency graph then the language is infinite

Example: $S \rightarrow AB$

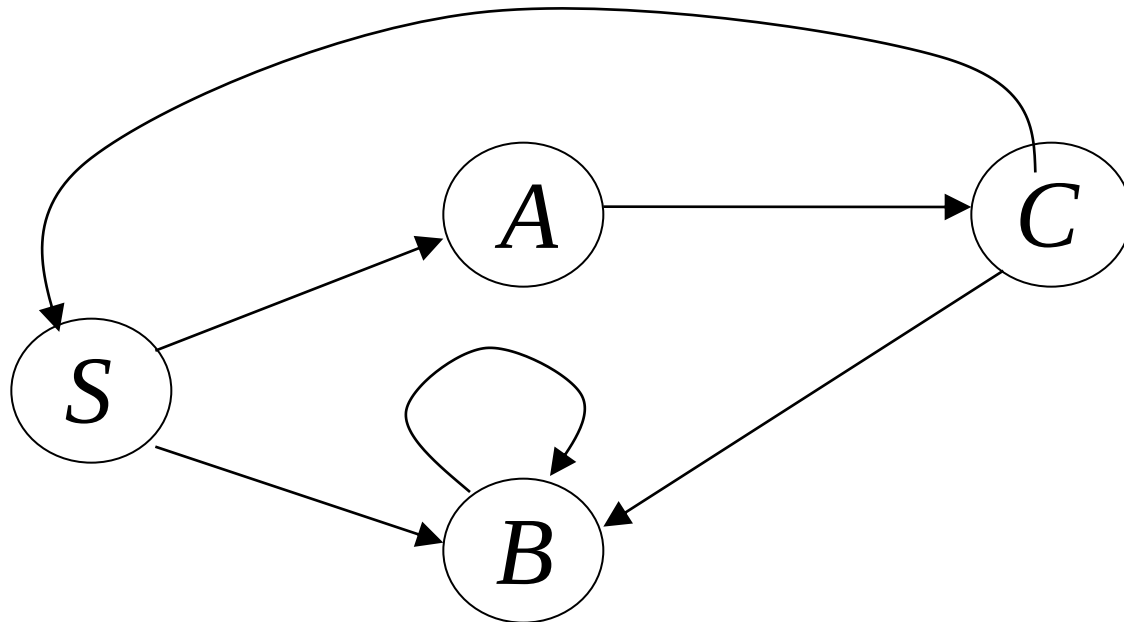
$A \rightarrow aCb \mid a$

$B \rightarrow bB \mid bb$

$C \rightarrow cBS$

Dependency graph

Infinite language



$$S \rightarrow AB$$

$$A \rightarrow aCb \mid a$$

$$B \rightarrow bB \mid bb$$

$$C \rightarrow cBS$$

$$S \Rightarrow AB \Rightarrow aCbB \Rightarrow acBSbB \Rightarrow acbbSbbb$$

$$S \overset{*}{\Rightarrow} acbbSbbb \overset{*}{\Rightarrow} (acbb)^2 S (bbb)^2$$

$$\overset{*}{\Rightarrow} (acbb)^i S (bbb)^i$$